Converting fractions to terminating decimals

1 Decimals that have a finite number of digits after the decimal point are called terminating decimals.

2 Without using a calculator, work out the decimal value of $\frac{7}{8}$. 0.875

3 Select all the fractions that can be written as terminating decimals. (Tick **3** correct answers)

- [x] $\frac{7}{16}$
- [x] $\frac{7}{5}$
- [ ] $\frac{7}{15}$
- [x] $\frac{7}{20}$
- [ ] $\frac{7}{30}$

4 Sofia uses her calculator to convert 0.3756 to a fraction. She gets an answer of $\frac{939}{\square}$.

What number should go in the square? 2500

5 Match each fraction to its terminating decimal. You can use a calculator for this question. (Write the correct letter in each box)

a	$\frac{82}{125}$
b	$\frac{3281}{5000}$
c	$\frac{21}{32}$
d	$\frac{26}{40}$
e	$\frac{180}{300}$

a	0.656
c	0.65625
e	0.6
d	0.65
b	0.6562

6 Lucas says that $\frac{66}{1320}$ is not equivalent to a terminating decimal. Without using a calculator, explain whether Lucas is correct or not. (Tick **1** correct answer)

- [ ] Lucas is correct as 1320 has factors of 3 and 11
- [ ] Lucas is correct as only fractions with a denominator of 10 terminate
- [x] Lucas is wrong as the fraction simplifies to $\frac{1}{20}$ which terminates
- [ ] Lucas is wrong as the denominator is a multiple of 10
- [ ] Lucas is wrong as 66 is a multiple of 2